

SIMATS ENGINEERING

ASSESSMENT TOOL 3

Web Development Standards Implementation

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Introduction

Modern web development follows internationally accepted standards to ensure that websites are secure, accessible, maintainable, and compatible with different browsers and devices. HTML provides the structure of web pages, HTTP defines communication between browsers and servers, XML stores and transfers structured data, and web protocols enable clients and servers to exchange information. In this scenario, a startup is developing a modern web application and faces issues related to HTML elements, HTTP methods, XML syntax, and communication protocols. The following sections identify the correct solutions and explain why they are appropriate while discussing the limitations of the incorrect options.

1. HTML Video Embedding

Correct Answer: B. `<video src="video.mp4" controls></video>`

The HTML5 `<video>` element is the standard method for embedding video content directly into a webpage. It eliminates the need for external plugins such as Adobe Flash and works efficiently across modern browsers. The `controls` attribute provides playback controls including play, pause, volume, progress bar, and fullscreen mode. Developers can also provide multiple `<source>` elements to support different video formats such as MP4, WebM, and Ogg. Captions can be added using the `<track>` element to improve accessibility.

Advantages include browser compatibility, responsive media support, improved accessibility, and better user experience. Invalid tags such as `<media>`, `<movie>`, and `<vid>` are not recognized by HTML standards and therefore cannot embed videos correctly.

2. HTTP Method for Form Submission

Correct Answer: POST

HTTP defines several request methods, each serving a different purpose. For a registration form, POST is the correct method because it sends form data in the request body instead of appending it to the URL. This makes POST suitable for sensitive information such as usernames, passwords, phone numbers, and email addresses. POST also supports larger amounts of data and is commonly used to create new resources on the server.

GET is mainly used to retrieve information and exposes parameters in the URL. PUT is intended for updating existing resources, while DELETE removes resources from the server. Although POST does not encrypt data by itself, it becomes secure when used with HTTPS.

3. Semantic Navigation Element

Correct Answer: <nav>

Semantic HTML improves both accessibility and search engine optimization. The <nav> element clearly identifies a section of a webpage containing navigation links such as menus, sidebars, or breadcrumbs. Screen readers can easily identify navigation areas, making websites more accessible to users with disabilities. Search engines also understand the page structure better.

Using generic elements like <div> for navigation makes the code less meaningful. Although <header> may contain navigation links, its main purpose is introductory content. <section> represents a thematic grouping of content rather than navigation.

4. XML Syntax

Correct Answer:

```
<note>  
  <to>User</to>  
  <from>Admin</from>  
</note>
```

XML requires documents to be well-formed. Every opening tag must have a matching closing tag, tags must be properly nested, attribute values must be enclosed in quotation marks, and only one root element should exist. The given correct example satisfies all these rules.

Option A is incomplete because the closing `</note>` tag is missing. Option C contains mismatched tags where `<to>` is incorrectly closed with `</from>`. Option D contains an incomplete closing tag because the final angle bracket is missing. Such syntax errors prevent XML parsers from reading the document correctly.

5. Web Communication Protocol

Correct Answer: HTTP

HTTP (Hypertext Transfer Protocol) is the foundation of communication on the World Wide Web. It allows web browsers to send requests to web servers and receive responses containing HTML pages, images, videos, and other resources. Today, HTTPS is widely used because it encrypts communication using SSL/TLS, protecting user information during transmission.

FTP is designed for transferring files, SMTP is used for sending emails, and SNMP is used for monitoring and managing network devices. Therefore, HTTP is the appropriate protocol for general web communication.

Conclusion

The successful development of modern web applications depends on following web standards.

The HTML5 <video> element provides efficient multimedia support, POST securely submits user data, the <nav> element improves accessibility and SEO, well-formed XML ensures reliable data exchange, and HTTP/HTTPS enables communication between browsers and servers.

Adhering to these standards results in secure, scalable, user-friendly, and maintainable web applications.

Summary Table

Topic	Correct Answer	Reason
Video	<video>	Standard HTML5 tag
Form	POST	Secure submission
Navigation	<nav>	Semantic navigation
XML	Option B	Well-formed XML
Protocol	HTTP	Web communication